5.3.3 LU factorization with partial pivoting

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Week 05 due Feb 12, 2025 20:19 IST

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The whole point of talking about pivot matrices was to have an elegant way of

describing what we're going to call LU factorization with partial pivoting. And

we're going to describe a specific algorithm for computing that.

Namely

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Reading assignment

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Read Unit 5.3.3 of the notes. [LINK]

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? HW 5.3.3.1 hyperlink text error
You may want to use Assignments/Week05/matlab/test_LU_right_looking.m to c... 2

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going to show that

actually this right here,

because these are Gaus transforms, and the very special Gauss transforms, all of

this can be rearranged so that all of the pivoting happens up front. And then

new Gauss transforms L_0 star, L_1 star and so forth L_n -1 star, end up to

the left of matrix U. And then if you multiply all of those together, in the

exact same way that we did before where we just place the multipliers in the

right spot in the matrix

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Homework 5.3.3.1

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Implement the algorithm

$[A, p] = \text{LUpiv-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right), p \rightarrow \left(\begin{array}{c} p_T \\ \hline p_B \end{array}\right)$

A_{TL} is 0×0, p_T has 0 elements

while $n(A_{TL}) < n(A)$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right), \left(\begin{array}{c} p_T \\ \hline p_B \end{array}\right) \rightarrow \left(\begin{array}{c} p_0 \\ \hline \pi_1 \\ \hline p_2 \end{array}\right)$$

$\pi_1 := \text{maxi}\left(\begin{array}{c} \alpha_{11} \\ \hline a_{21} \end{array}\right)$

$$\left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right) := \left(\begin{array}{c|c} I & 0 \\ \hline 0 & P(\pi_1) \end{array}\right) \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right)$$

$a_{21} := a_{21}/\alpha_{11}$

$A_{22} := A_{22} - a_{21}a_{12}^T$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array}\right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array}\right), \left(\begin{array}{c} p_T \\ \hline p_B \end{array}\right) \leftarrow \left(\begin{array}{c} p_0 \\ \pi_1 \\ \hline p_2 \end{array}\right)$$

endwhile

as

```
function [ A_out, p_out ] = LUpiv_right_looking( A )
```

by completing the code in Assignments/Week05/matlab/LUpiv_right_looking.m. Input is an $m \times n$ matrix A. Output is the matrix A that has been overwritten by the LU factorization and pivot vector p. You may want to use Assignments/Week05/matlab/test_LUpiv_right_looking.m to check your implementation.

The following utility functions may come in handy:

- Assignments/Week05/matlab/maxi.m.
- Assignments/Week05/matlab/Swap.m

which we hope are self explanatory.

- [x] done

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